Truncation Cross Entropy Loss for Remote Sensing Image Captioning

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***Abstract*— Recently, remote sensing image captioning (RSIC) has drawn an increasing attention. In this field, the encoder–decoder-based methods have become the mainstream due to their excellent performance. In the encoder–decoder framework, the convolutional neural network (CNN) is used to encode a remote sensing image into a semantic feature vector, and a sequence model such as long short-term memory (LSTM) is subsequently adopted to generate a content-related caption based on the feature vector. During the traditional training stage, the probability of the target word at each time step is forcibly optimized to 1 by the cross entropy (CE) loss. However, because of the variability and ambiguity of possible image captions, the target word could be replaced by other words like its synonyms, and therefore, such an optimization strategy would result in the overfitting of the network. In this article, we explore the overfitting phenomenon in the RSIC caused by CE loss and correspondingly propose a new truncation cross entropy (TCE) loss, aiming to alleviate the overfitting problem. In order to verify the effectiveness of the proposed approach, extensive comparison experiments are performed on three public RSIC data sets, including UCM-captions, Sydney-captions, and RSICD. The state-of-the-art result of Sydney-captions and RSICD and the competitive results of UCM-captions achieved by TCE loss demonstrate that the proposed method is beneficial to RSIC.**

***Index Terms*— Image captioning, overfitting, remote sensing, truncation cross entropy (TCE) loss.**

I. Introduction

HIGH-resolution remote sensing images, which have a wide range of applications [1]–[7], can be easily obtained nowadays due to the rapid development of the remote sensing technology. How to mine efficiently the relationship between the visual features and the semantic information hidden in the remote sensing images has been widely concerned. Motivated by natural image captioning [8]–[14], remote sensing image captioning (RSIC) [15] has been explored in the past few years. RSIC, which combines computer vision with natural language processing [16], aims to let machine automatically generate human understandably descriptions from the given remote sensing images. Benefiting from the technology of deep learning, the methods based on the neural encoder–decoder architecture have gradually become a growing trend in the field of RSIC. This kind of methods typically contains two stages: image understanding and caption generation. In the image-understanding phase, the convolutional neural network (CNN) is used as a feature extractor to encode the input remote sensing image into a high-level semantic feature vector of fixed dimension, which aggregates the visual features and objects of the images. Subsequently, in the caption-generation phase, the logic and syntax relationship of these visual features and objects is decoded into a well-formed sentence by a sequence model, such as long short-term memory (LSTM) [17]. In this way, many concise and meaningful sentences would be generated for the input remote sensing images, which is in line with the logic of human language and is helpful for human cognitive understanding. Commonly, during the training stage, the encoder–decoder models are optimized by the cross entropy (CE) loss [18], which regards the word prediction as a classification task [19]. At each time step, a word is predicted by the decoder network according to the following three components: the feature vector, the previous word, and the generated sentence. It is worth mentioning that the probability of the target word at each time step is forcibly optimized to 1 by the CE loss. Nevertheless, due to the variability and ambiguity of the possible image captions, the target word may be strongly related to other nontarget words like its synonyms. For example, as shown in Fig. 1, the given remote sensing image can be described by several sentences of Fig. 1(a)–(d). For Fig. 1(a) and (b), the object words of "plants" and "trees" could be replaced by each other. Similarly, for Fig. 1(c) and (d), "Many" and "Some" are the alternatives. To improve the output probability of the target word, the unreasonable noise in the image would be learned. Thus, such 1-probability optimization strategy would lead to the overfitting of the network when the noise is learned. In order to explore quantitatively the overfitting caused by the CE loss, in this article, three different forms of CE loss are defined first according to the optimized objects, which are positive cross entropy (PCE) loss, negative cross entropy (NCE) loss, and positive–negative cross entropy (PNCE) loss. In the PCE loss, only the target word regarded as the positive sample is adopted to optimize the model, and it is intrinsically equal to

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(a) Many tennis courts arranged neatly with some plants surrounded.
(b) Many tennis courts arranged neatly with some trees surrounded.
(c) Many tennis courts arranged neatly with some plants surrounded.
(d) Some tennis courts arranged neatly with some plants surrounded.

Fig. 1. Sentences for an image is not unique, and some words can be replaced by other words like their synonyms.

the normal CE loss used in RSIC. In contrast, when using the NCE loss, only the nontarget words treated as negative samples are employed for the optimization of the model. In terms of PNCE loss, both the positive and negative samples (i.e., all the words in the vocabulary) would be adopted to optimize the whole model.

In addition, to alleviate the overfitting problem, a novel truncation cross entropy (TCE) loss for the remote sensing image-caption generation is first presented in this article. Different from the conventional CE loss, the proposed TCE loss is a piecewise loss that consists of two parts: a traditional CE loss and a truncation loss. More specifically, during the training stage, an upper limit would be set to $1-\gamma$ to decide which loss would be selected. When the output probability of the current word is lower than $1-\gamma$, the traditional CE loss would be employed to optimize the whole model. In addition, when the output probability of the word exceeds the upper limit of $1-\gamma$, the truncation loss mechanism would be activated immediately, which means the target probability of this word will not be optimized higher than $1-\gamma$. In this way, a margin of γ, which is also quite valuable for the sentence, can be reserved for the nontarget words in the vocabulary. Such a fuzzy mechanism can effectively alleviate the overfitting phenomenon caused by the CE loss and enhance the performance of the base model. Overall, the main contributions of this article can be summarized as follows.

1) To explore the overfitting caused by the CE loss in RSIC, we present three types of losses, including PCE loss, NCE loss, and PNCE loss according to the optimized words. Moreover, the corresponding comparison experiments are conducted with quantitative analysis.
2) A novel TCE loss, which aims to alleviate the overfitting problem caused by CE loss, is first proposed for the caption generation of the remote sensing images. By reserving a probability margin for the nontarget words, the proposed TCE loss is helpful to generate more flexible and concise descriptions for the remote sensing images and further enhance the generalization performance of the whole model.
3) Different CNNs combined with LSTM are applied on three public data sets to verify the effectiveness of the proposed TCE loss. The state-of-the-art result of Sydney-captions and RSICD and the competitive results of UCM-captions, which are achieved by the TCE loss, demonstrate that the proposed approach is beneficial to RSIC.

The remainder of this article is organized as follows. Section II introduces the related works of RSIC and CE loss. In Section III, we describe the proposed TCE loss-based method in detail. The experimental results and analysis on three data sets are introduced in Section IV. Finally, the conclusions are provided in Section V.

II. Related Work

In this section, the relevant work of RSIC and CE loss will be briefly introduced.

A. Remote Sensing Image Captioning

In general, the methods of RSIC can be roughly divided into three categories: retrieval-based methods, template-based methods, and encoder–decoder model-based methods.

Retrieval-based methods depend on the retrieval results and the matching degree. For example, Wang *et al.* [20] presented a collective semantic metric learning architecture to describe the image content more diversely. Based on the technology of metric learning, this model maps the dimensions of input-image representation and their corresponding caption representation to the same space. By computing the distances between the test image and all collective captions, the caption with the smallest distance would be picked up as the final descriptive sentence. However, since the generated captions are searched from the existing database, the methods based on retrieval are difficult to perform well when facing an input image that has low similarity with all the images in the database.

Template-based methods aim to define a fixed sentence template with several blanks reserved. Then, after extracting the features from the input images, the detected objects, their attributes, and the relationship among them would be correspondingly filled in the reserved blanks. For instance, a fully convolutional network (FCN)-based method was proposed by Shi and Zou [21] for better describing the remote sensing images in human language, where the FCNs [22] are employed to capture the elements with three different levels in a remote sensing image. In addition, a collection of triplets is used to guide the generation of the captions. Nevertheless, for the template-based methods, one problem researchers have to consider is that the predefined template has greatly limited the flexibility of the generated captions, thus making the forms of the generated sentences quite rigid.

The encoder–decoder model-based methods [23] are end to end, which mainly contains two stages of encoding and decoding. The purpose of encoding is to represent the input image as a feature vector of fixed dimension, where different kinds of feature extractors, especially CNN, can be employed. For decoding, a sequence model such as recurrent neural network (RNN) [24] or LSTM [17] can be used to generate the corresponding caption word by word with the guidance of the feature vector. The methods of this type for RSIC were first proposed in [25] by Qu *et al.*, where a multimodal neural network model is especially designed to understand the remote sensing images in the semantic level. In addition, different CNNs combined with the RNN or LSTM are explored to

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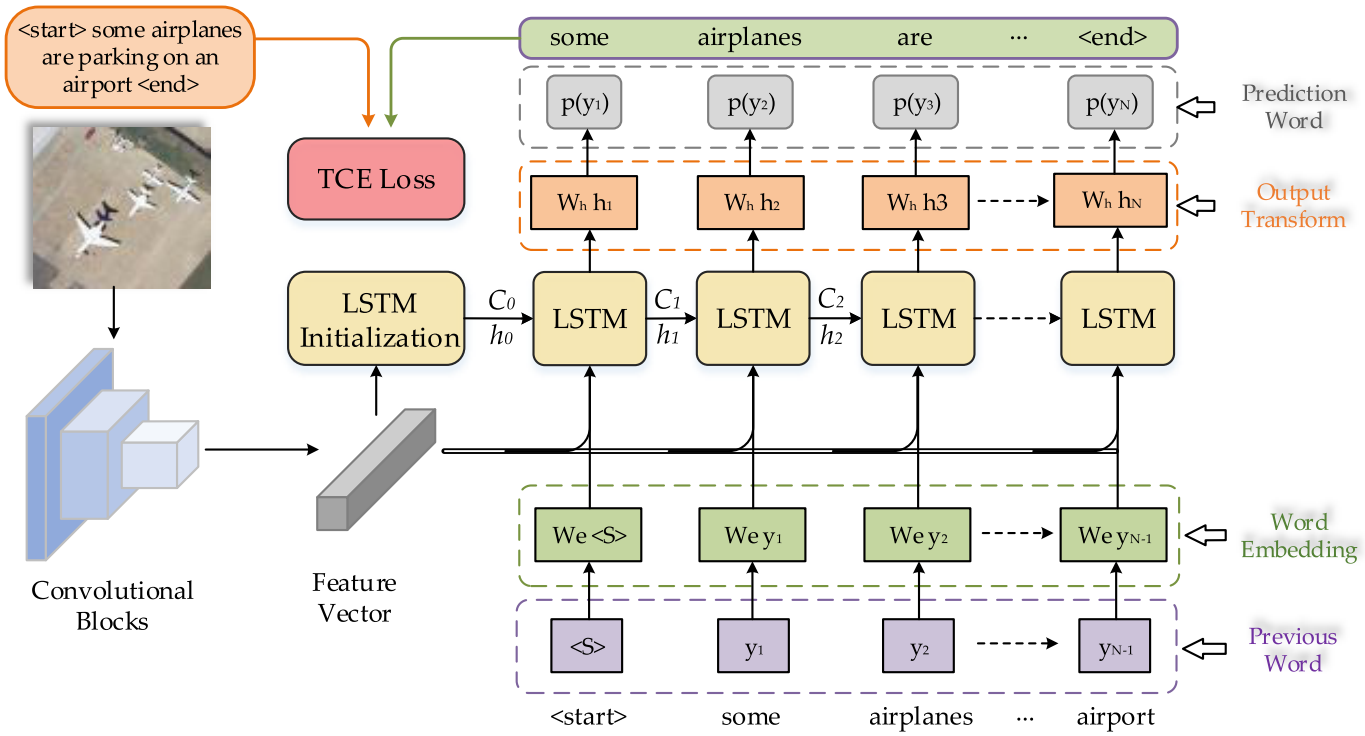


Fig. 2. Framework of the proposed TCE loss-based method.

find the best combination for RSIC in this article with two public data sets released. After that, Lu *et al.* [26] presented a new public data set for RSIC, and both the "soft" and "hard" attentions are introduced in this article. After that, the attention mechanism is widely used in the field of RSIC. For example, Yuan *et al.* [27] presented a multilevel attention module, concentrating on different spatial positions and different scales. In general, methods based on the encoder–decoder model have become more and more popular because of their excellent performance.

B. Cross Entropy Loss for RSIC

CE is mainly used to indicate the difference information between two probability distributions. In general, for the encoder–decoder-based model of RSIC, the procedure of word prediction can be viewed as a classification task, where CE loss is employed for optimization. For the decoding part, all words in the candidate vocabulary would be assigned a probability after going through a softmax layer. The goal of CE loss is to optimize the probability of the target word to 1 while the nontarget words to 0, and only one word would be generated at each time step. The CE loss used in RSIC can be formulated as

$$\mathcal{L}_{\mathrm{CE}} = -\sum_{i=1}^{N} y^{(i)} * \log \hat{y}^{(i)} \tag{1}$$

where $\mathcal{L}_{\mathrm{CE}}$ denotes the CE loss, and $y^{(i)}$ and $\hat{y}^{(i)}$ refer to the ground-truth label and its factual output probability, respectively. N represents the total number of classes.

III. Methodology

In this section, the proposed TCE loss is introduced in detail and used to optimize the captioning models obeying the classical encoder–decoder framework for RSIC. As shown in Fig. 2, the workflow of the TCE loss-based encoder–decoder model mainly consists of three components: 1) feature extractor; 2) caption generator; 3) TCE loss.

A. Feature Extractor

For an encoder–decoder-based method, the goal of image representation is to encode the input images into high-level semantic features, which is a vital part for RSIC. Traditionally, the methods of feature extraction can be generally divided into two categories: handcrafted features and deep learning features. Nowadays, a large number of deep learning models, especially CNN [28]–[31], have shown surprising feature representation in a wide range of image fields, and they also work well in RSIC. Different from the methods based on the handcrafted features, which require considerable engineering skills and domain knowledge, the CNN can automatically learn features from data through a deep-structured neural network. In addition, since many processing layers are generally contained, the CNN can learn and obtain more powerful feature representations with multiple levels of abstraction. Since the CNN has achieved excellent visual image representation, it is used to extract the features of the remote sensing images. For a CNN model, it is usually composed of the stacked convolutional blocks (backbone) and fully connected layer (classifier). However, the classifier is redundant for RSIC. Therefore, the last fully connected layer is removed from the

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CNN and the remaining CNN backbone is used as the feature extractor. To verify that the proposed TCE loss is not limited by the CNN models, in this article, several different kinds of CNN models are used for feature extraction.

Given an RGB remote sensing image of I, the multichannel semantic feature map with the spatial size of $H \times W$, which is denoted as $F \in \mathbb{R}^{C\times H\times W}$, is extracted from the image by the CNN backbone. Here, C is the number of feature channels. It is formulated as

$$F = \mathrm{CNN}_{\mathrm{conv}}(I) \tag{2}$$

where $\mathrm{CNN}_{\mathrm{conv}}$ represents the CNN backbone. To reduce the model parameters, F is converted into the corresponding feature vector of $v \in \mathbb{R}^C$ by a global average pooling (GAP) layer [32], which is formulated as

$$v = \mathrm{GAP}(F). \tag{3}$$

B. Caption Generator

In general, the goal of caption generation is to decode the feature vector extracted by the CNN into a sequence of words. Here, LSTM [17], which is a widely used sequence generation model, is used as a caption generator in our model. Different from the traditional sequence models, since LSTM is able to store the long-term memory information, it has a good ability to solve the problem of gradient vanishing.

The structure of LSTM used in this article is shown in Fig. 3, and it is constructed by several basic blocks stacked together. It is worth noting that the core of LSTM is the cell state, where the transfer process of information from the last cell state c_{t-1} to the current cell state c_t is specially controlled by three gates, i.e., input gate i_t, forget gate f_t, and output gate o_t. At time step t, the update procedure of LSTM can be represented as follows:

$$i_t = \sigma(W_{xi}x_t + W_{hi}h_{t-1} + b_i) \tag{4}$$
$$f_t = \sigma\left(W_{xf}x_t + W_{hf}h_{t-1} + b_f\right) \tag{5}$$
$$\tilde{c_t} = \tanh(W_{xc}x_t + W_{hc}h_{t-1} + b_i) \tag{6}$$
$$c_t = f_t * c_{t-1} + i_t * \tilde{c_t} \tag{7}$$
$$o_t = \sigma(W_{xo}x_t + W_{ho}h_{t-1} + b_o) \tag{8}$$
$$h_t = o_t * \tanh(c_t) \tag{9}$$

where h_t represents the hidden state of the LSTM at time t, which is also the output of LSTM at this time step. tanh and σ sigmoid are, respectively, the hyperbolic tangent function and sigmoid function. All the W and b are the learnable parameters of weights and bias. In addition, x_t denotes the input of LSTM at time t, and the definition of x_t is formulated by

$$x_t = \mathrm{concat}(v, y_{t-1}) \tag{10}$$

where v and y_{t-1} denote the feature vector extracted by the CNN and the output of LSTM at time $t-1$, respectively. x_t is the combination of v and y_{t-1}.

In general, the overall procedure of decoding can be denoted as

$$h_t = \mathrm{LSTM}(x_t) \tag{11}$$
$$y_t = W_{ed} * h_t \tag{12}$$

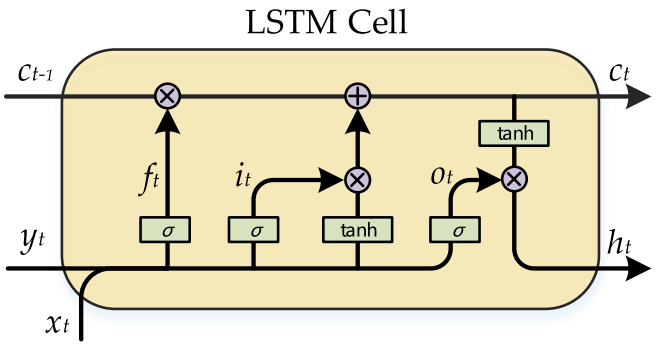


Fig. 3. Structure of LSTM.

where y_t represents the output of an word-embedding operation on h_t. At each time step, only one word would be output by LSTM.

C. Optimization Loss

1) Loss Definition: In image captioning, the word prediction is regarded as a multiclassification task. Therefore, the word probability distribution of the generated sentence and reference sentence can be denoted as $\hat{\boldsymbol{y}} = [\hat{y}_1, \ldots, \hat{y}_N]$, $\hat{y}_t \in \mathbb{R}^K$, and $\boldsymbol{y} = [y_1, \ldots, y_N]$, $y_t \in \mathbb{R}^K$. Here, N is the sentence length and K is the vocabulary size. For a label sequence of $\boldsymbol{s} = [s_1, \ldots, s_N]$, $s_i \in \mathbb{R}^1$ refers to the word number in the vocabulary at time step t. $\hat{\boldsymbol{y}}$ is the one-hot version of $\boldsymbol{s}$. In general, for a predicted sentence, the loss used for optimization during training can be denoted as

$$\mathcal{L} = -\sum_{t=1}^{T} \mathcal{L}_t \tag{13}$$

where t and T represent the current time step and the total time steps, respectively.

2) PCE, NCE, and PNCE Losses: At time step t, the subloss of $\mathcal{L}_t$ can be expressed in three different forms of PCE loss (denoted as $\mathcal{L}_{\mathrm{PCE}}$), NCE loss (denoted as $\mathcal{L}_{\mathrm{NCE}}$), and PNCE loss, which aim at optimizing the target words directly and indirectly. It is worth mentioning that the concepts of NCE loss and PNCE loss are first proposed in this article.

In $\mathcal{L}_{\mathrm{PCE}}$, only the target word, which can be regarded as the positive sample, is treated as the optimization objective. It is defined as $\mathcal{L}_{\mathrm{PCE}}$:

$$\begin{aligned}\mathcal{L}_{\mathrm{PCE}} &= -y_t^{(s_t)} * \log \hat{y}_t^{(s_t)} \\ &= -\log \hat{y}_t^{(s_t)}\end{aligned} \tag{14}$$

where s_t denotes the number of the target word at time step t. $y_t^{(s_t)}$ and $\hat{y}_t^{(s_t)}$, respectively, refer to the output probability and target probability of s_t. It is obvious that the value of $y_t^{(s_t)}$ is equal to 1. The goal of PCE loss is to optimize the output probability of the target word to 1. PCE loss intrinsically is the typical CE loss [33] when it is used in RSIC.

In L_{NCE}, only the nontarget words at time step t, which can be considered as the negative samples in contrast to the

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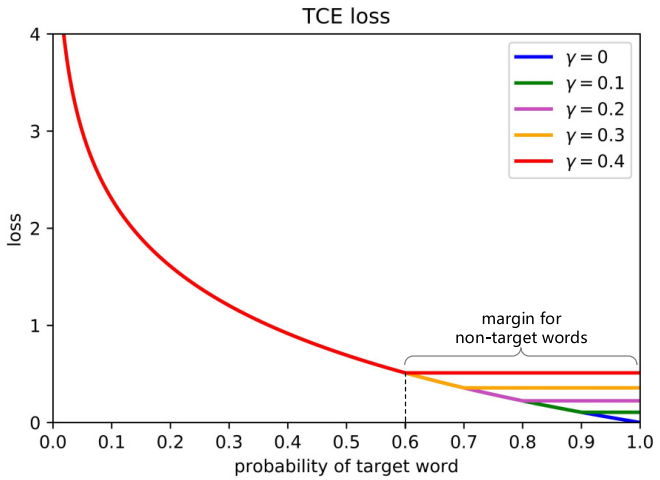


Fig. 4. TCE loss with different values of γ.

positive sample, are treated as the optimization objectives. It is defined as $\mathcal{L}_{\mathrm{NCE}}$

$$\mathcal{L}_{\mathrm{NCE}} = -\frac{1}{N-1}\sum_{i\neq s_t}^{N}\left(1-y_t^{(i)}\right)*\left(1-\log\hat{y}_t^{(i)}\right) = -\frac{1}{N-1}\sum_{i\neq s_t}^{N}\left(1-\log\hat{y}_t^{(i)}\right) \tag{15}$$

where $y_t^{(i)}$ and $\hat{y}_t^{(i)}$, respectively, refer to the output probability and target probability of the ith word in the vocabulary except s_t. The goal of NCE loss is to optimize the sum of the output probability of all the nontarget words to 0.

Although both PCE and NCE loss have the same goal to make the output probability of s_t converge to 1, their optimization objectives and strategies are different. For more comprehensive ablation study, they are composed of a new combination loss of the PNCE loss denoted as L_{PNCE}, which is defined as

$$\mathcal{L}_{\mathrm{PNCE}} = \mathcal{L}_{\mathrm{PCE}} + \mathcal{L}_{\mathrm{NCE}} = -\log\hat{y}_t^{(s_t)} - \frac{1}{N-1}\sum_{i\neq s_t}^{N}\left(1-\log\hat{y}_t^{(i)}\right). \tag{16}$$

3) TCE Loss: The aim of TCE loss is designed to alleviate the overfitting problem of the captioning model for the RSIC at the training stage. The proposed TCE loss denoted as $\mathcal{L}_{\mathrm{TCE}}$ is a piecewise function, which is made up of two components: a common PCE loss and a truncation loss. It is formulated as

$$\mathcal{L}_{\mathrm{TCE}} = \begin{cases} -y_t^{(s_t)} * \log\hat{y}_t^{(s_t)}, & \text{if } y_t^{(s_t)} < 1-\gamma \\ -\log(1-\gamma), & \text{otherwise} \end{cases} \tag{17}$$

where γ is the value of the truncation threshold that reserves a margin for the nontarget words during the training stage. In this article, γ is set to 0, 0.1, 0.2, 0.3, and 0.4, respectively. It is worth mentioning that TCE loss is actually the PCE/CE loss when γ is equal to 0. $y_t^{(s_t)}$ and $\hat{y}_t^{(s_t)}$, respectively, refer to the output probability and target probability of s_t. The proposed TCE loss is illustrated in Fig. 4.



1. There are many residential areas near the school.
2. A playground is built next to a white building.
3. Several playgrounds are in the big school.
4. Several playgrounds are in the big school.
5. There are many residential areas near the school.

Fig. 5. Example of an image and its corresponding five captions in RSICD.

TCE loss is developed based on the PCE loss. Different from the PCE loss, however, an upper limit of $1-\gamma$ is set in the TCE loss to determine which loss would be chosen during the optimization procedure. Supposing that the output probability of the target word (i.e., $y_t^{(s_t)}$) exceeds the upper limit of $1-\gamma$, the loss function switches from PCE loss to Truncation loss. Therefore, $y_t^{(s_t)}$ would be optimized to a value of $1-\gamma$ instead of 1, which means a probability margin of γ would be especially reserved for the remaining nontarget words in the vocabulary except the target word. In this way, the overfitting phenomenon caused by the common CE loss can be effectively alleviated.

IV. Experiments

In this section, experiments are conducted on three data sets to verify the effectiveness and generalization of the proposed method. First, we introduce the experimental data sets and evaluation metrics of RSIC. Then, the experimental settings are provided in detail. Following that, the overfitting phenomenon caused by CE loss is discussed, and the ablation experiments of the PCE, NCE, PNCE, and TCE losses are performed. Finally, our results are compared with some state-of-the-art models with comparative analysis.

A. Data Sets

There are three widely used RSIC data sets of different sizes, including Sydney-captions, UCM-captions, and RSICD.

1) Sydney-Captions: The Sydney-captions data set is proposed by Qu *et al.* [25], which is based on the Sydney data set [34]. It contains 613 images with seven scene categories, including industrial, rivers, residential, meadow, runway, airport, and ocean. All the images are collected from Google Earth of Sydney, Australia. Moreover, the resolution of each image is 0.5 m. For each image, five reference sentences are given to abstract the content from different observers. Totally, there are 237 different words in the Sydney-captions data set. Eighty percentage of the images in Sydney-captions are used for training, 10% for validation, and the rest 10% for test.

2) UCM-Captions: The UCM-captions data set [25] is also proposed in [25], which is based on the UC Merced (UCM) land-use data set [35]. It contains 2100 high-resolution remote sensing images with 21 scene categories, including building, beach, airplane, chaparral, forest, harbor, freeway, overpass, intersection, runway, river, agricultural, dense residential, tennis court, sparse residential, golf course, baseball diamond, medium residential, parking lot, mobile home park, and storage tank. All the images are measuring 256 × 256 pixels with a pixel resolution of 0.3048 m. There are totally 368 different

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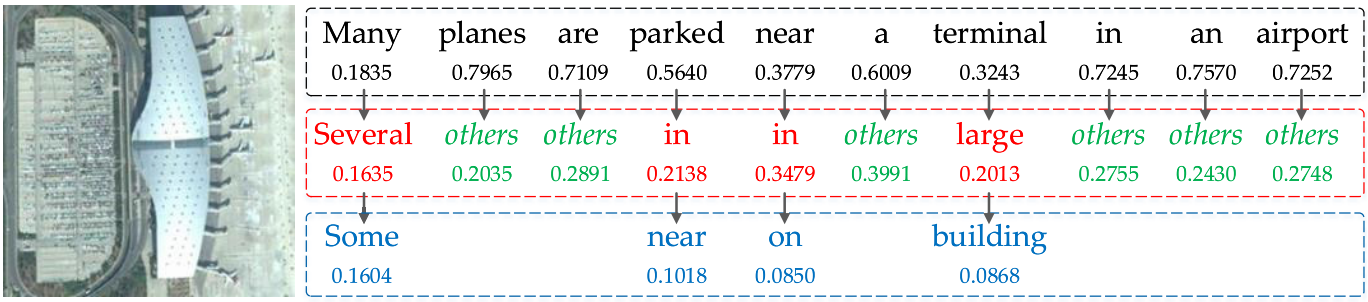


Fig. 6. Example of the probability of the generated top-three words at each time step. The first, second, and third rows show the words of the **maximal**, the **submaximal**, and the **third** probabilities at each time step. In addition, the ***others*** denotes all the nonmaximal words in the vocabulary.

words in the UCM-captions data set. Similar to the Sydney-captions data set, five descriptions are given for each image. In addition, the splitting ratio is the same as Sydney-captions.

3) RSICD: The RSICD data set is proposed by Lu *et al.* [26], which consists of 10921 images measuring 224 × 224 pixels. All the images are collected from MapABC, Baidu Map, Google Earth, and Tianditu with different resolutions. Totally, there are 3323 different words in the RSICD data set. Similar to the previous data sets, five reference sentences are provided for each image. It is mentioning that there are repetitions in the five reference sentences, and its splitting ratio is the same as Sydney-captions and UCM-captions. There is a captioning example of a remote sensing image, as shown in Fig. 5.

B. Evaluation Metrics

In the field of remote sensing image-caption generation, four evaluation metrics are commonly used, including BLEU [40], METEOR [41], ROUGE_L [42], and CIDEr [43]. The value ranges of BLEU, METEOR, and ROUGE_L are all from 0 to 1, and the value range of CIDEr is from 0 to 5. For all four metrics, the larger the value is, the better the quality of the generated captions is.

1) BLEU: BiLingual Evaluation Understudy (BLEU) [40] was first used to evaluate the quality of the machine translation models, and now, it is widely used in various sequence-generation tasks. By calculating the precision of n-gram of different lengths and performing geometric weighted average, BLEU aims to measure the n-gram coincidence between the generated sequence and the reference sequence. Here, the value of n is set to 1, 2, 3, and 4 corresponding to BLEU1, BLEU2, BLEU3, and BLEU4.

2) METEOR: Metric for Evaluation of Translation with Explicit ORdering (METEOR) [41] is measured by computing an alignment between the generated sentence and the reference sentence. Based on a single-precision weighted harmonic mean and single-word recall rate, METEOR considers both precision and recall rate; thus, it can solve some of the defects inherent to the BLEU standard.

3) ROUGE_L: Recall-Oriented Understudy for Gisting Evaluation (ROUGE) [42] was first applied in the field of text summarization. It is similar to BLEU, while the difference is that ROUGE_L concentrates on calculating the recall rate. F-measure based on the longest common subsequence (LCS) is adopted in the ROUGE_L metric to evaluate the similarity of the reference sentence and the generated sentence.

4) CIDEr: Consensus-based Image Description Evaluation (CIDEr) [43] is a metric for image captioning. In CIDEr, each sentence is translated into a "document" and expressed as a term frequency inverse document frequency (TF-IDF) vector. Then, the cosine similarity between the reference sentence and the generated sentence would be calculated by the model. Compared with the abovementioned metrics, CIDEr considers the frequency of the words in the vocabulary.

C. Experimental Settings

In this article, all the experiments are built on Pytorch 1.3 of Python 3.7. Four different CNN models pretrained on ImageNet, including AlexNet [44], VGG16 [45], ResNet18 [46], and GoogleNet [47], are employed to explore the efficient CNN feature extractor for RSIC. We remove all layers after the last convolutional layer of the CNNs, which are replaced by a GAP layer. The output of the CNN extractors is a high-dimension feature vector, which is the input of a one-layer LSTM. For AlexNet, the dimension of the feature vector is 256, while for the other three CNN extractors, the dimension of the feature vector is 512. During the decoder stage, the dimension of both the word embedding and hidden state of LSTM is set to 512 in all the experiments. In order to improve the memory efficiency, the data are processed in batches with the batch size set to 64. During the whole training phase, Adam is used to optimize the models. All the models are trained for 50 epochs with the learning rate of 1e−4. For the TCE loss, we set the truncation threshold value γ to 0, 0.1, 0.2, 0.3, and 0.4, respectively, to explore the effect of margin value.

D. Exploring the Efficient CNN Extractor

In order to explore the efficient CNN extractor for RSIC, four classical CNN architectures are adopted here, including AlexNet, VGG16, ResNet18, and GoogleNet. AlexNet [44] is designed by implementing the deep CNN structure for the first time in the large-scale image data sets. Compared with AlexNet, VGG16 [45] is improved by replacing the large convolution kernels with several small consecutive convolution kernels, with much deeper layers to learn more complex patterns. GoogleNet [47] has a unique inception structure consisting of several convolutional kernels of different sizes,

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TABLE I
EXPERIMENTAL RESULTS WITH DIFFERENT CNN EXTRACTORS ON THE UCM-CAPTIONS DATA SET

CNN Extractor	BLEU1	BLEU2	BLEU3	BLEU4	METEOR	ROUGE_L	CIDEr
AlexNet	0.7567	0.6728	0.6123	0.5626	0.3984	0.6755	2.4720
VGG16	0.7758	0.6965	0.6410	0.5929	0.4275	0.7073	2.6150
ResNet18	0.8079	0.7384	0.6861	0.6397	0.4496	0.7306	2.7688
GoogleNet	**0.8193**	**0.7522**	**0.7007**	**0.6559**	**0.4708**	**0.7483**	**2.8996**

TABLE II
EXPERIMENTAL RESULTS OF PCE (I.E., CE), NCE, AND PNCE LOSSES ON THREE DATA SETS

Dataset	Loss	BLEU1	BLEU2	BLEU3	BLEU4	METEOR	ROUGE_L	CIDEr
UCM-captions	$\mathcal{L}_{PCE}$	**0.8193**	**0.7522**	**0.7007**	**0.6559**	**0.4708**	**0.7483**	2.8996
	$\mathcal{L}_{NCE}$	0.8034	0.7327	0.6775	0.6283	0.4586	0.7382	2.8390
	$\mathcal{L}_{PNCE}$	0.8035	0.7442	0.6943	0.6474	0.4617	0.7433	**2.9470**
Sydney-captions	$\mathcal{L}_{PCE}$	**0.7960**	**0.7238**	**0.6496**	**0.5810**	**0.4244**	**0.7034**	2.0180
	$\mathcal{L}_{NCE}$	0.7751	0.6882	0.6066	0.5346	0.4015	0.6837	1.9248
	$\mathcal{L}_{PNCE}$	0.7877	0.7106	0.6512	0.6015	0.4147	0.6920	**2.1999**
RSICD	$\mathcal{L}_{PCE}$	**0.7440**	**0.6183**	**0.5295**	**0.4615**	**0.3421**	**0.6500**	**2.3524**
	$\mathcal{L}_{NCE}$	0.7011	0.5608	0.4710	0.4065	0.3059	0.6068	1.9541
	$\mathcal{L}_{PNCE}$	0.7290	0.6058	0.5203	0.4555	0.3248	0.6438	2.2900

TABLE III
EXPERIMENTAL RESULTS OF TCE LOSS WITH DIFFERENT MARGIN VALUES ON THREE DATA SETS. WHEN γ IS EQUAL TO 0, TCE LOSS IS EQUIVALENT TO PCE/CE LOSS

Dataset	Threshold	BLEU1	BLEU2	BLEU3	BLEU4	METEOR	ROUGE_L	CIDEr
UCM-captions	$\gamma = 0$	0.8193	0.7522	0.7007	0.6559	0.4708	0.7483	2.8996
	$\gamma = 0.1$	0.8210	**0.7622**	**0.7140**	**0.6700**	**0.4775**	0.7567	2.8547
	$\gamma = 0.2$	0.8237	0.7604	0.7089	0.6681	0.4762	**0.7591**	**2.9809**
	$\gamma = 0.3$	0.8057	0.7415	0.6880	0.6387	0.4694	0.7451	2.9038
	$\gamma = 0.4$	**0.8242**	0.7592	0.7059	0.6569	0.4730	0.7551	2.9332
Sydney-captions	$\gamma = 0$	0.7960	0.7238	0.6496	0.5810	0.4244	0.7034	2.0180
	$\gamma = 0.1$	0.7937	0.7304	**0.6717**	**0.6193**	**0.4430**	**0.7130**	**2.4042**
	$\gamma = 0.2$	0.7873	0.7173	0.6512	0.5926	0.4429	0.7039	2.1447
	$\gamma = 0.3$	**0.8067**	**0.7356**	0.6677	0.6082	0.4266	0.7050	2.3607
	$\gamma = 0.4$	0.7820	0.7062	0.6355	0.5757	0.4332	0.6991	2.0517
RSICD	$\gamma = 0$	0.7440	0.6183	0.5295	0.4615	0.3421	0.6500	2.3524
	$\gamma = 0.1$	0.7589	0.6264	0.5319	0.4597	0.3431	0.6565	2.3237
	$\gamma = 0.2$	0.7438	0.6138	0.5212	0.4513	0.3353	0.6448	2.3041
	$\gamma = 0.3$	**0.7608**	**0.6358**	**0.5471**	**0.4791**	0.3425	**0.6687**	**2.4665**
	$\gamma = 0.4$	0.7512	0.6270	0.5398	0.4735	**0.3439**	0.6548	2.4143

and thus is better for the multiscale feature-extraction task. For taking advantage of a novel residual block, ResNet [46] is designed to alleviate the problem of gradient disappearance in deep networks with the skip-connection operation, which makes the network training faster and more stable. Since the number of images in the UCM-captions data set is moderate, experiments with different CNN extractors are conducted on UCM-captions to find the efficient CNN extractor for RSIC.

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TABLE IV
COMPARISON RESULTS OF DIFFERENT METHODS ON THE UCM-CAPTIONS DATA SET

	BLEU1	BLEU2	BLEU3	BLEU4	METEOR	ROUGE_L	CIDEr
FC-Att+LSTM [36]	0.8135	0.7502	0.6849	0.6352	0.4173	0.7504	2.9958
SM-Att+LSTM [36]	0.8154	0.7575	0.6936	0.6458	0.4240	0.7632	3.1864
Soft Attention [26]	0.7454	0.6545	0.5855	0.5250	0.3886	0.7237	2.6124
Hard Attention [26]	0.8157	0.7312	0.6702	0.6182	0.4263	0.7698	2.9947
GCN Based Multi-Level Attention [27]	**0.8330**	**0.7712**	**0.7154**	0.6623	0.4371	0.7763	3.1684
sound-a-a [37]	0.7484	0.6837	0.6310	0.5896	0.3623	0.6579	2.7281
RTRMN(statistical) [38]	0.8028	0.7322	0.6821	0.6393	0.4258	0.7726	3.1270
VAA [39]	0.8192	0.7511	0.6927	0.6387	0.4380	**0.7824**	**3.3946**
The Proposed Method	0.8210	0.7622	0.7140	**0.6700**	**0.4775**	0.7567	2.8547

TABLE V
COMPARISON RESULTS OF DIFFERENT METHODS ON THE SYDNEY-CAPTIONS DATA SET

	BLEU1	BLEU2	BLEU3	BLEU4	METEOR	ROUGE_L	CIDEr
FC-Att+LSTM [36]	0.8076	0.7160	0.6276	0.5544	0.4099	0.7114	2.2033
SM-Att+LSTM [36]	0.8143	0.7351	0.6586	0.5806	0.4111	0.7195	2.3021
GoogleNet Soft Attention [26]	0.7322	0.6674	0.6223	0.5820	0.3942	0.7127	**2.4993**
GoogleNet Hard Attention [26]	0.7591	0.6610	0.5889	0.5258	0.3898	0.7189	2.1819
GCN Based Multi-Level Attention [27]	**0.8233**	**0.7548**	0.6587	0.6003	0.4202	**0.7237**	2.3110
sound-a-a [37]	0.7093	0.6228	0.5393	0.4602	0.3121	0.5974	1.7477
VAA [39]	0.7431	0.6646	0.6029	0.5495	0.3930	0.6999	2.4073
The Proposed Method	0.7937	0.7304	**0.6717**	**0.6193**	**0.4430**	0.7130	2.4042

TABLE VI
COMPARISON RESULTS OF DIFFERENT METHODS ON THE RSICD DATA SET

	BLEU1	BLEU2	BLEU3	BLEU4	METEOR	ROUGE_L	CIDEr
FC-Att+LSTM [36]	0.7459	0.6250	0.5338	0.4574	0.3395	0.6333	2.3664
SM-Att+LSTM [36]	0.7571	0.6336	0.5385	0.4612	0.3513	0.6458	2.3563
Soft Attention [26]	0.6753	0.5308	0.4333	0.3617	0.3255	0.6109	1.9643
Hard Attention [26]	0.6669	0.5182	0.4164	0.3407	0.3201	0.6084	1.7925
GCN Based Multi-Level Attention [27]	0.7597	**0.6421**	**0.5517**	0.4623	**0.3543**	0.6563	2.3614
sound-a-a [37]	0.6196	0.4819	0.3902	0.3195	0.2733	0.5143	1.6386
RTRMN(statistical) [38]	0.6102	0.4514	0.3535	0.2859	0.2751	0.5452	1.4820
The Proposed Method	**0.7608**	0.6358	0.5471	**0.4791**	0.3425	**0.6687**	**2.4665**

The experimental results with four pretrained CNN extractors on the UCM-captions data set are provided in Table I, where the truncation threshold value of γ is fixed to 0, i.e., PCE loss. In the experiments, GoogleNet achieves the best performance in all kinds of evaluation metrics. Taking CIDEr, for example, GooleNet has an increase of 0.4276 as compared with AlexNet, 0.2846 as compared with VGG16, and 0.1308 as compared with ResNet18. Thus, it would be used as the feature extractor in all the subsequent experiments. It is probable that GoogleNet can fuse multiscale features by applying several convolution kernels of different sizes at each layer, which is suitable for the feature extraction of the remote sensing images.

E. Results of PCE, NCE, and PNCE Losses

To quantificationally explore the influence of the positive and negative sample training strategies for RSIC, we use the PCE loss (i.e., CE loss), NCE loss, and PNCE loss to optimize the same encoder–decoder model (GoogleNet + LSTM) on three data sets. The model optimized by PCE loss only uses the positive samples for optimization during the training stage. Similarly, all the negative samples are used to optimize the model for the NCE loss. When it comes to PNCE loss, both the positive and negative samples are adopted for optimization.

The experimental results are shown in Table II. Overall, the encoder–decoder model optimized by the PCE loss achieves the best performance on all three data sets, and the PNCE loss has the suboptimal results and NCE loss performs worst. Taking RSICD for instance, the BLEU1, BLEU2, BLEU3, BLEU4, METEOR, ROUGE_L, and CIDEr scores of the model optimized by PCE loss are, respectively, 0.7440, 0.6183, 0.5295, 0.4615, 0.3421, 0.6500, and 2.3524, which have increases of 0.0429, 0.0575, 0.0585, 0.055, 0.0362, 0.0432, and 0.3983, respectively, as compared with the model

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optimized by NCE loss. When NCE is added to PCE loss, i.e., PNCE loss, the scores drop by, respectively, 0.015, 0.0125, 0.0092, 0.006, 0.0173, 0.0062, and 0.0624 as compared with the single PCE loss. Such results show that compared with the PCE loss, the NCE loss plays a negative role in improving the caption quality.

First, it should be noted that the model would be overfitting and further decrease the performance when the output probability of all the words is optimized to 1. Although the training targets of the PCE, NCE, and PNCE losses are the same to make the output probability of the words at each time optimized to 1, they have different training effects. NCE loss optimizes the negative words at each time step, while the PCE loss directly optimizes the positive words. Limited by the implicit optimization targets, it is understandable that the performance of the NCE loss falls behind the PCE loss. However, when NCE is added to PCE loss, the new PNCE loss would lead to the faster convergence of the model and make the model miss the better parameter space.

F. Results of TCE Loss

In order to slow down the speed toward overfitting during the training stage and further search for the parameters with more generalization performance, we use TCE loss to optimize the same encoder–decoder-based model (GoogleNet + LSTM) on three data sets. Different from the PCE, NCE, and PNCE losses mentioned above, the output probability of the target word would not be optimized to 1 by the TCE loss, which means a probability margin could be reserved for the remaining nontarget words related except the target word. The truncation threshold denoted as γ is set to 0, 0.1, 0.2, 0.3, and 0.4, respectively, to explore further the influence of probability margin in this section.

Table III shows the comparative results of PCE loss and TCE loss with different margin values on three data sets. Compared with absolute PCE loss, when adding margin max operation (TCE loss), the performance of RSIC obtains a comprehensive enhancement. Similarly, taking RSICD, for example, the BLEU1, BLEU2, BLEU3, BLEU4, METEOR, ROUGE_L, and CIDEr scores are, respectively, 0.7608, 0.6358, 0.5471, 0.4791, 0.3425, 0.6687, and 2.4665 under the circumstance that γ is set to 0.3. Compared with the PCE loss-optimized model (namely, when γ is equal to 0 in Table III), there are wide increases of 0.0168, 0.0175, 0.0176, 0.0176, 0.0004, 0.0187, and 0.1141, respectively. In addition, it is worth mentioning that the optimal value of γ is 0.1 for both UCM-captions and Sydney-captions, and it increases to 0.3 when using the largest data set of RSICD. Overall, according to the results, it can be observed that by setting a truncation threshold, the model optimized by TCE loss achieves better performance than the PCE loss, and the best threshold varies with the data sets.

It is notable that the best optimal margin value of γ increases with the size of a data set, more specifically, the size of its vocabulary. Since RSICD contains more words (the total number of words is 1187) than the other two data sets, the same features or objects in RSICD can be described with more synonymous words than them in the other two data sets. Therefore, the problem of overfitting between the interchangeable words during the training stage is more serious for it. Furthermore, more probability margins of each word need to be reserved for its relative words in RSICD, which means that the optimal truncation threshold of γ should be bigger. In contrast, the number of words in UCM-captions and Sydney-captions is, respectively, 368 and 237; thus, a small value of γ can perform well for them.

In summary, compared with the pure PCE loss, by reserving a suitable probability margin for the relative words to replace the target word, TCE loss is rather effective to alleviate the overfitting for all three data sets during the optimization process.

G. Comparison With the State-of-the-Art Methods

To verify the effectiveness of the proposed method, comparison experiments are conducted on several state-of-the-art approaches, including FC-Att+LSTM [36], SM-Att+LSTM [36], Soft Attention [26], Hard Attention [26], GCN-Based Multi-Level Attention [27], sound-a-a [37], RTRMN(statistical) [38], and VAA [39]. All these methods are built on the encoder–decoder framework. Both FC-Att+LSTM and SM-Att+LSTM are proposed in [36], where the multiple attributes, which are extracted from the high-level semantic features of the remote sensing images by the attention mechanism, are used to obtain better caption quality based on the basic model. The main difference in these two models is that their high-level attributes are the output of the different layers of the CNN (the last fully connected layer for FC-Att+LSTM, while the softmax layer for SM-Att+LSTM). Soft attention and hard attention are introduced in [26]. In particular, soft attention is a deterministic method, where a weight is given to decide which part of the image should be paid more attention. Differently, hard attention is a stochastic method, where a sampling strategy is employed to concentrate on different parts of the image and then reinforcement learning is adopted to better improve the performance of the model. GCN-based multi-level attention is proposed by Yuan *et al.* [27]. Typically, a multi-level attention module is presented for better performance of RSIC, and the position-adaptive and scale-adaptive image representations can be learned by this model. The method of sound-a-a is proposed [37], where a sound mechanism is introduced as an active attention to improve the quality of caption generation. RTRMN(statistical) is proposed in [38]. This model aims to overcome the drawback of the long-term information dilution in the RNN, and a topic word strategy is presented to use fully the given five reference captions. VAA is proposed in [39], where a novel visual aligning attention model is presented to address the problem of not explicitly training the attention layers in the encoder–decoder model.

1) Results on UCM-Captions: Table IV shows the comparison results between the aforementioned methods and the proposed CNN-LSTM model with TCE loss on the UCM-captions data set. It can be seen that the results of the proposed method are the best among all the methods in terms

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Fig. 7. Examples of test images and the corresponding generated captions. (a) There are two tennis courts arranged neatly and surrounded by some plants. (b) There are two white storage tanks on the ground. (c) An overpass goes across the roads with some cars on the roads. (d) Many boats docked in lines at the harbor and the water is deep blue. (e) A residential area with many houses arranged neatly and divided into rectangles by some roads. (f) Many buildings and some green trees are in a commercial area. (g) Many airplanes are parked in an airport. (h) A baseball field is near several buildings and some green trees. (i) Some buildings and green trees are in two sides of a river with a bridge. (j) Some green trees and several buildings are around a circle building. (k) Many cars are parked in a parking lot. (l) Many buildings and green trees are around a football field.

of BLEU4 and METEOR. According to BLEU1, BLEU2, BLEU3, ROUGE_L, and CIDEr, the performance of the proposed method just slightly falls behind the other state-of-the-art approaches.

2) Results on Sydney-Captions: The comparative results on the Sydney-captions data set are shown in Table V. It can be observed that the proposed method performs best in terms of BLEU3, BLEU4, and METEOR. As for BLEU1, BLEU2, ROUGE_L, and CIDEr, our method achieves a competitive performance compared with other state-of-the-art approaches.

3) Results on RSICD: As the biggest RSIC data set, RSICD contains much more images than the other two data sets with complex scenarios, and thus, its result is more robust and convincing. The comparative results are shown in Table VI. The highest score for each metric is marked in bold. It can be observed that compared with all the existing state-of-the-art approaches, the proposed method takes the first place in BLEU1, BLUE4, ROUGE_L, and CIDEr, and the competitive results in BLEU2, BLEU3, and METEOR. Such experimental results completely prove the effectiveness and superiority of our approach.

4) Advantages of Our Method: According to the analysis of the methods and comparative experimental results on the three data sets, it can be found that compared with the other state-of-the-art methods, our approach has two obvious advantages, which can be summarized as follows:

Without adding extra modules or parameters: Different from the other comparison methods, the proposed method aims to improve the encoder–decoder-based model from the aspect of training optimization, which means the model of our method is the basic CNN combined with LSTM, without adding any extra modules. In other words, there are no extra parameters embedded into our model, so the computational complexity and training cost of TCE loss is almost equal to the basic model, which operates much faster than the aforementioned methods.

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Getting better performance: From Tables IV to VI, the comparative results show that the proposed method achieves superior performance on the RSIC data sets. Typically, when the number of n-grams for the BLEU metric (from BLEU1 to BLEU4) increases, our method gets relatively better performance than others. It is probably caused by that the constraints on each word in phrase would be relaxed during the training stage, which makes our method more flexible for multiple consecutive words. It is beneficial to the open caption from different observers. Some examples of test images and the corresponding generated captions are shown in Figs. 6 and 7.

V. Conclusion

In this article, we first quantificationally explore the overfitting phenomena in RSIC caused by the CE loss. To deal with this problem, a novel TCE loss is especially proposed for RSIC. By setting a truncation threshold, the output probability of the target word would not be optimized to 1 by TCE loss, and thus, a probability margin can be reserved for the remaining words in the vocabulary. Based on a classic encoder–decoder model (CNN and LSTM), comparative experiments are conducted on three widely used RSIC data sets, including UCM-captions, Sydney-captions, and RSICD. The superior performance of the experimental results compared with other state-of-the-art methods shows that the proposed TCE loss is rather effective for RSIC.

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